

Chenyang LIU

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EDUCATION

Duke University

PhD in Computer Science, advised by Prof. Matthew Lentz and Prof. Kartik Nayak

Durham, USA

Aug. 2024 – Now

Shanghai Jiao Tong University

B.Sc. in Information Security; GPA: 90.33/100 Rank: 3/86

Member of Zhiyuan Honors Program of Engineering in Zhiyuan College; Top 5%

Shanghai, China

Sep. 2020 – June 2024

RESEARCH INTEREST

Confidential Computing and Blockchain.

RESEARCH EXPERIENCES

Duke System Group

Initial Project advised by Prof. Matthew Lentz and Prof. Kartik Nayak

Durham, USA

Aug. 2024 – Nov. 2025

Decentralized Proposer-as-a-Service

- **Removing Relays in Ethereum PBS:** Removed centralized relays and outsourced the combined role of proposer and relays to a group of TEEs (PEs)
- **Security-in-Depth Design:** Employed $(5f - 1)$ -psync-VBB protocol with secret share and threshold signature to achieve fair exchange in Ethereum PBS

NSEC Lab

Undergraduate Research Assistant advised by Prof. Guoxing Chen

Shanghai, China

Feb. 2023 – present

Closing Cache Side Channels via Minimal Cache Fingerprints

- **Memory Footprint Minimization:** Designed a software-simulated and fully-associated cache architecture to hide access pattern, enhancing resistance to access-pattern based attacks
- **Compiler Modification:** Instrumented enclave code into 64-bytes blocks by modifying LLVM backend
- **Verification of the OS:** Proposed a novel contract to verify the OS under SGX's threat model

Formally verified privileged enclave as a lightweight and efficient confidential computing platform

- **Case Study and Evaluation:** Helped implemented three case studies and evaluated the performance.

PUBLICATIONS

A Formal Approach to Multi-Layered Privileges for Enclaves

G Yang, C Liu, Z Huang, G Chen, H Fu, Y Zhang, H Zhu

NDSS 2025

DPaaS: Improving Decentralization by Removing Relays in Ethereum PBS

C Liu, I Abraham, M Lentz, K, Nayak

In Submission

TEACHING SERVICES

Teaching Assistant

Undergraduate Course: Introduction to Operating System

Duke University

Spring 2025

Teaching Assistant

Graduate Course: Operating System

Duke University

Fall 2025

SELECTED HONORS & AWARDS

Zhiyuan Honor Scholarship: Awards to students in Zhiyuan Honor Program (Top 5%)

2021 – 2023

First Class Network Security Scholarship: Awards to students who contribute to the research of security

2023

Academic Progress Scholarship: Awards to students who have made great academic progress in one year (3/88) 2022